

Nose and Paranasal Sinuses

Chronic sinusitis with nasal polyps (ALOX15, TSLP, IL33, HLA), allergic rhinitis and the mucosal barrier, and smell (olfactory receptors)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F's genes (genotyping chip + local imputation). It is not a diagnosis or a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Key variants by topic	6
5. GWAS loci burden	9
6. Favourable	10
7. Unfavourable	11
8. Mechanisms of ageing	12
9. Practical suggestions	13
10. Limitations	14
11. Methods and sources	15
12. Annex: all variants analysed	16

1. Summary

Unfavourable / to watch. Chronic sinusitis with nasal polyps is the weak point of the nose. The most recent genomic nasal polyp score (1.06 million variants) is at percentile **98.3** and the burden of 90 loci for chronic sinusitis and polyps at **86.4**, which confirms the 96th percentile in the Allergies, Asthma and Atopy report. However, two other scores are at percentiles 32.4 and **39.3**, and the consensus of the scores with good coverage is 39.3: the signal is mixed, with the balance tilting towards risk. In the genes: **ALOX15** G/G (without the rare protective allele), **TSLP** C/C (2 alleles), **GIGYF2** A/A (2 alleles) and 1 allele each in IL33, GPX4, TBX1, BACH2, IRF4 and LTC4S.

Favourable. Allergic rhinitis has a low predisposition: the burden of 376 loci for rhinitis and hay fever is at percentile **18.0** and the allergic disease score (asthma, hay fever or eczema) at percentile 21.5. The nasal polyp **HLA-DQA1** locus lacks the risk allele (A/A), and so does **EMSY/LRRC32** (G/G).

Smell (no judgement). The olfactory receptors carry no severe loss-of-function variants. OR5A1 has 2 copies of the β -ionone sensitivity allele, OR6A2 has 1 copy of the "soapy coriander" allele and OR7D4 has 1 copy of the allele that reduces the perception of androstenone. **UGT2A1/UGT2A2** has 1 copy of the allele that increases loss of smell during viral infections (T/C).

What it means. The nose has a profile of **type 2 inflammation of the sinus mucosa** (TSLP, IL33, ALOX15), with a higher-than-average risk of polyps, without the classic allergic background (allergic rhinitis unlikely). In practice, it is worth recognising the signs early: persistent nasal congestion, loss of smell and post-nasal drip for more than 12 weeks. Loss of smell is often the first sign of polyps.

2. How to read this report

- **Data:** genotypes from Sérgio F's chip (GRCh38, + strand) and, when the SNP is not on the chip, locally imputed genotypes (statistical imputation, public panel of 30x whole genomes) with quality DR2 ≥ 0.8 . Below this threshold the SNP is treated as not found in the data.
- **Key variants:** 21 SNPs selected from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 from the chip, 9 imputed and 0 not found.
- **Strand:** all genotypes are on the + strand of GRCh38. The published effect allele is compared with the observed alleles; 2 SNPs needed conversion to the complementary strand and 1 is palindromic (A/T or C/G), resolved by the reference allele and the position. Where the literature uses the gene strand, the letters may differ from those in this report.
- **Polygenic scores (PGS):** 9 PGS Catalog scores calculated locally with 1,571,835 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentile ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **GWAS loci burden:** for each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in the GWAS Catalog (1 row per SNP, the most significant) are added up as a simple count of effect alleles (each allele counts 1). Sérgio F's sum is compared with that of the reference population ($n = 633$) using exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium with each other, so the burden is a complement to, not a substitute for, the polygenic scores. 521 loci, 469 with genotype.
- **Judgement:** each result is classed as Favourable or Unfavourable (or typical / no judgement). A genotype is a probability, not a destiny; almost all effects are small and add to lifestyle.
- **Evidence:** Strong = replicated and with a clear clinical effect; Moderate = consistent GWAS/meta-analyses, modest effect; Weak = small or purely mechanistic studies.

3. Polygenic scores

Consensus by trait. For each group, the consensus is the **median of the percentiles of the scores with effect-size coverage $\geq 75\%$** , all calibrated in the same reference population ($n = 633$). The genomic column is the median of the scores with $\geq 100,000$ variants. Scores with coverage $< 75\%$ are shown in the tables but do not enter the consensus.

Group	Consensus (percentile)	Scores with coverage $\geq 75\%$	Genomic scores (percentile)	Genomic scores
Chronic sinusitis and nasal polyps	39.3	3 of 4	65.3	2
Allergic rhinitis and atopy	21.5	1 of 5	—	0

In all these scores, a higher percentile means more risk. The 2021 allergic rhinitis scores have 48–54% coverage and are left out of the consensus; the burden of 376 rhinitis loci is the best reading of that trait. For nasal polyps, the most recent genomic score (2024) disagrees with the 2022 scores.

Chronic sinusitis and nasal polyps

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Nasal polyps (genomic)	PGS004465 (Jung 2024, 1.06 million variants)	1,025,639 of 1,059,939 (coverage 96%)	98.3	MORE LIKELY	unfavourable
Nasal polyps (genomic)	PGS002060 (Privé 2022, 553,000 variants)	542,959 of 553,242 (coverage 97%)	32.4	TYPICAL	typical
Nasal polyps	PGS001848 (Privé 2022, 1,783 variants)	1,724 of 1,783 (coverage 97%)	39.3	TYPICAL	typical
Nasal polyps	PGS000934 (Tanigawa 2022, 129 variants)	63 of 128 (coverage 51%)	73.1	TYPICAL	typical

Allergic rhinitis and atopy

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Allergic rhinitis and hay fever	PGS001259 (Tanigawa 2022, 349 variants)	167 of 348 (coverage 48%)	72.4	TYPICAL	typical
Allergic rhinitis (diagnosed)	PGS001286 (Tanigawa 2022, 466 variants)	231 of 466 (coverage 54%)	63.7	TYPICAL	typical
Vasomotor and allergic rhinitis	PGS001109 (Tanigawa 2022, 1,910 variants)	894 of 1,907 (coverage 49%)	60.7	TYPICAL	typical
Allergic disease (asthma, hay fever or eczema)	PGS000810 (PGS Catalog 2021, 136 variants)	126 of 136 (coverage 91%)	21.5	TYPICAL	typical
General atopic disease	PGS003458 (Al-Janabi 2023, 124 variants)	32 of 124 (coverage 23%)	23.5	TYPICAL	typical

4. Key variants by topic

Replicated loci for chronic sinusitis with nasal polyps (epithelial alarmins, lipoxygenases and HLA), allergic rhinitis and atopy, and olfactory receptors with a known effect. Published alleles with a protective effect were inverted to the risk direction.

Chronic sinusitis and nasal polyps

Type 2 inflammation of the sinus mucosa: epithelial alarmins (TSLP, IL33), 15-lipoxygenase and leukotrienes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ALOX15	rs34210653	T560M	G/G	chip	G (2 copies)	unfavourable	T560M of 15-lipoxygenase: the rare A allele (loss of function) strongly protects against nasal polyps; the common G allele is the risk allele [GWAS Catalog: Nasal polyps, allele G, OR 3.12, p=8E-27, PMID 30643255]	Strong
TSLP	rs1837253		C/C	chip	C (2 copies)	unfavourable	TSLP (epithelial alarmin): nasal polyps, asthma and atopy [GWAS Catalog: Nasal polyps, allele C, beta 0.279 [0.23-0.32] unit increase, p=2E-32, PMID 34594039]	Strong
IL33	rs1888909		T/C	imputed DR2=1.00	T (1 copy)	unfavourable (mild)	IL33 (type 2 alarmin): nasal polyps [GWAS Catalog: Nasal polyps, allele T, OR 1.34, p=7E-32, PMID 30643255]	Strong
HLA-DQA1	rs1391371		A/A	imputed DR2=1.00	T (0 copies)	favourable	HLA-DQA1: nasal polyps (~1.5×) [GWAS Catalog: Nasal polyps, allele T, OR 1.47, p=7E-49, PMID 30643255]	Strong
GIGYF2	rs6704768		A/A	chip	A (2 copies)	unfavourable	GIGYF2: chronic sinusitis with polyps [GWAS Catalog: Chronic rhinosinusitis with nasal polyposis, allele A, OR 1.13, p=7E-17, PMID 41213931]	Moderate
GPX4	rs4807542		G/A	chip	A (1 copy)	unfavourable (mild)	GPX4 (glutathione peroxidase 4, defence against ferroptosis): chronic sinusitis with polyps [GWAS Catalog: Chronic rhinosinusitis with nasal polyposis, allele A, OR 1.15, p=9E-14, PMID 41213931]	Moderate
TBX1	rs1978060		A/G	chip	A (1 copy)	unfavourable (mild)	TBX1 (development of the face and sinus): chronic sinusitis with polyps [GWAS Catalog: Chronic rhinosinusitis with nasal polyposis, allele A, OR 1.12, p=1E-13, PMID 41213931]	Moderate
BACH2	rs62408224		A/G	imputed DR2=1.00	G (1 copy)	unfavourable (mild)	BACH2 (regulation of lymphocytes): chronic sinusitis with polyps [GWAS Catalog: Chronic rhinosinusitis with nasal polyposis, allele G, OR 1.15, p=1E-17, PMID 41213931]	Moderate
IRF4	rs1050979		G/A	imputed DR2=1.00	A (1 copy)	unfavourable (mild)	IRF4: chronic sinusitis with polyps [GWAS Catalog: Chronic rhinosinusitis with nasal polyposis, allele A, OR 1.11, p=2E-13, PMID 41213931]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LTC4S	rs730012		A/C	chip	C (1 copy)	unfavourable (mild)	–444A/C of LTC4S: the C allele is associated with more cysteinyl leukotrienes (aspirin-intolerance pathway)	Weak

Allergic rhinitis and mucosal barrier

Genes of allergic sensitisation and innate immunity of the nasal mucosa.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
EMSY/LRRC32	rs2155219		G/G	chip	T (0 copies)	favourable	11q13.5 (EMSY/LRRC32): the T allele increases allergic rhinitis, asthma and eczema	Moderate
IL1RL1/IL18R1	rs10865050		G/G	imputed DR2=1.00	G (2 copies)	unfavourable	IL-33 receptor (ST2): allergic disease and hay fever [GWAS Catalog: Allergic disease (asthma, hay fever or eczema), allele G, OR 1.13, p=7E-61, PMID 29083406]	Moderate
TLR1	rs5743604		G/A	imputed DR2=1.00	A (1 copy)	unfavourable (mild)	TLR1/6/10 (innate immunity): hay fever and eczema [GWAS Catalog: Hay fever and/or eczema, allele A, OR 1.15, p=1E-72, PMID 31361310]	Moderate
D2HGDH	rs34290285		G/G	chip	G (2 copies)	unfavourable	D2HGDH (2-hydroxyglutarate metabolism): allergic disease [GWAS Catalog: Allergic disease (asthma, hay fever or eczema), allele G, OR 1.11, p=6E-58, PMID 31361310]	Moderate
WDR36/TSLP	rs1438673		C/T	chip	C (1 copy)	unfavourable (mild)	WDR36/TSLP: allergic disease [GWAS Catalog: Allergic disease (asthma, hay fever or eczema), allele C, OR 1.09, p=7E-62, PMID 31361310]	Moderate
CLEC16A	rs12935657		G/G	imputed DR2=0.98	G (2 copies)	unfavourable	CLEC16A: allergic disease and autoimmunity [GWAS Catalog: Allergic disease (asthma, hay fever or eczema), allele G, OR 1.09, p=6E-45, PMID 31361310]	Weak

Smell (olfactory receptors)

Variants that change the perception of specific odours; no effect on health (no judgement).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
OR6A2	rs72921001		C/A	chip	A (1 copy)	no judgement	OR6A2 (aldehyde receptor): the A allele is associated with perceiving coriander as soapy	Moderate
OR5A1	rs6591536	D183N	A/A	chip	A (2 copies)	no judgement	OR5A1 D183N: the A allele (Asn) gives much more sensitivity to β -ionone (violet scent)	Moderate
OR7D4	rs61729907	R88W	G/A	chip	T (1 copy)	no judgement	OR7D4 R88W: the T allele reduces the perception of androstenone (smell of sweat and pork)	Moderate
OR2J3	rs28757581		A/A	imputed DR2=1.00	T (2 copies)	no judgement	OR2J3 (T113A): perception of cis-3-hexen-1-ol (cut-grass smell)	Weak

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
UGT2A1/ UGT2A2	rs7688383	T/C		imputed DR2=0.98	T (1 copy)	unfavourable (mild)	UGT2A1/UGT2A2 (olfactory epithelium enzymes): loss of smell and taste in viral respiratory infections [GWAS Catalog: COVID-19 symptoms (loss of smell or taste vs no loss of smell, allele T, OR 1.12, p=4E-15, PMID 35039640]	Moderate

5. GWAS loci burden

Sum of GWAS Catalog loci ($p \leq 5 \times 10^{-8}$) for chronic sinusitis and nasal polyps, allergic rhinitis and odour identification. Alleles are oriented towards more risk (or, for smell, in the direction of the published effect) and the sum is compared with the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F's effect alleles	Calibrated variants	Percentile	Judgement
Chronic sinusitis and nasal polyps	90 (78)	86 of 156	78	86.4	unfavourable
Allergic rhinitis and hay fever	407 (376)	377 of 752	368	18.0	favourable
Smell (odour identification)	24 (15)	17 of 30	15	82.5	no judgement

6. Favourable

- **Allergic rhinitis and hay fever:** burden of 376 loci at percentile **18.0**; allergic disease at percentile 21.5. **Moderate.**
- **HLA-DQA1 without the nasal polyp risk allele (A/A) and EMSY/LRRC32 without the atopy allele (G/G).** **Moderate.**
- **Olfactory receptors without severe loss-of-function variants. Moderate.**

7. Unfavourable

- **Chronic sinusitis with nasal polyps:** 2024 genomic score at percentile **98.3**; burden of 90 loci at **86.4**; consensus at percentile 39.3 (the 2022 scores are at percentiles 32.4 and 39.3). **Moderate.**
- **TSLP C/C** and **GIGYF2 A/A** (2 alleles each); **ALOX15 G/G** without the rare protective allele; 1 allele each in **IL33**, **GPX4**, **TBX1**, **BACH2** and **IRF4**. **Moderate.**
- **LTC4S -444C A/C:** 1 allele associated with more leukotrienes (the pathway of aspirin intolerance with polyps). **Weak.**
- **UGT2A1/UGT2A2 T/C:** 1 allele that increases loss of smell during viral infections by about 11%. **Moderate.**

8. Mechanisms of ageing

Mapping onto the 84 ageing mechanisms of Sérgio F's immortality goals (only the mechanisms touched by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	Type 2 inflammation of the sinus mucosa (TSLP, IL33, ALOX15); polyps with the genomic score at percentile 98.3	Unfavourable (mild)
Altered Intercellular Communication	Epithelial alarmins (TSLP, IL33) that activate type 2 innate lymphoid cells	Unfavourable (mild)
Immunosenescence	Allergic rhinitis is unlikely (loci at percentile 18.0); the systemic allergic response is low	Favourable
Oxidative Stress	GPX4 (defence against lipid peroxidation and epithelial ferroptosis) with 1 risk allele	Unfavourable (mild)
Neuroinflammation	The olfactory epithelium is a gateway to the brain; loss of smell is an early marker of neurodegeneration and frailty	Conditional

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Recognise chronic sinusitis early:** congestion, loss of smell or post-nasal drip for more than 12 weeks justifies an ear, nose and throat consultation. Loss of smell may be the first sign of polyps. Suggestion (not applied).
2. **Nasal rinsing with saline** during periods of congestion or exposure to dust and pollution; it is the baseline measure with the most evidence. Suggestion (not applied).
3. **Aspirin and NSAIDs:** if asthma or congestion ever appears after aspirin or ibuprofen, tell the doctor (aspirin-exacerbated respiratory disease, linked to leukotrienes and polyps). Suggestion (not applied).
4. **Smell training** if the sense of smell declines after a viral infection: 4 odours, 20 seconds each, twice a day, for months. Suggestion (not applied).
5. **Avoid nasal sprays or gels containing zinc** (they can cause permanent loss of smell). Suggestion (not applied).
6. **Already in the plan and protective:** omega-3 at percentile 3.2–4 g/day and vitamin D (both modulate type 2 inflammation), clean indoor air and no tobacco. Suggestion (not applied).

10. Limitations

- The nasal polyp scores disagree (from 32 to 98); the loci burden and the most recent score point to high risk, the 2022 scores to typical.
- The allergic rhinitis scores have about 50% coverage; the loci burden is the best reading.
- There are hundreds of olfactory receptors, almost all without a studied effect; the 4 read here describe specific odours only.
- Chronic sinusitis depends heavily on the environment (pollution, tobacco, infections, sinus anatomy).
- The loci burden adds up SNPs in linkage disequilibrium; it is descriptive.
- Conclusions from genes only; this is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F's chip (GRCh38, + strand). Missing SNPs come from local statistical imputation (public panel of 30x whole genomes), used only with $DR2 \geq 0.8$. No genetic data left the analysis computer.
- **Alleles and strand:** the published effect allele is compared with the alleles of the imputed VCF; for palindromic SNPs the + strand is assumed (the GWAS Catalog and PGS Catalog convention). Published alleles with a protective effect were inverted to the risk direction.
- **Polygenic scores:** GRCh38 harmonised files from the PGS Catalog (9 scores: nasal polyps, allergic rhinitis and atopy). Score = $\sum \text{coefficient} \times \text{dosage}$. The percentile is empirical in the reference population ($n = 633$), using the same variants. Consensus = median of the scores with effect-size coverage $\geq 75\%$.
- **Loci burden:** GWAS Catalog ($p \leq 5 \times 10^{-8}$), 3 sets, alleles oriented by the sign of the published effect, with empirical calibration in the same reference population.
- **Interpretation:** Kristjansson 2019 (ALOX15 and nasal polyps), Bohman 2017 and Ferreira 2017 (chronic sinusitis, rhinitis and atopy), Jaeger 2013 (OR5A1), Keller 2007 (OR7D4), Eriksson 2012 (OR6A2), Shelton 2022 (UGT2A1/UGT2A2), PGS Catalog, GWAS Catalog and ClinVar.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Strand	Interpretation
Chronic sinusitis and nasal polyps	ALOX15	rs34210653	G/G	chip	G	2	+ strand	2 copies of the risk allele
Chronic sinusitis and nasal polyps	TSLP	rs1837253	C/C	chip	C	2	+ strand	2 copies of the risk allele
Chronic sinusitis and nasal polyps	IL33	rs1888909	T/C	imputed DR2=1.00	T	1	+ strand	1 copy of the risk allele
Chronic sinusitis and nasal polyps	HLA-DQA1	rs1391371	A/A	imputed DR2=1.00	T	0	+ strand (palindromic; + assumed)	without the risk allele
Chronic sinusitis and nasal polyps	GIGYF2	rs6704768	A/A	chip	A	2	+ strand	2 copies of the risk allele
Chronic sinusitis and nasal polyps	GPX4	rs4807542	G/A	chip	A	1	+ strand	1 copy of the risk allele
Chronic sinusitis and nasal polyps	TBX1	rs1978060	A/G	chip	A	1	+ strand	1 copy of the risk allele
Chronic sinusitis and nasal polyps	BACH2	rs62408224	A/G	imputed DR2=1.00	G	1	+ strand	1 copy of the risk allele
Chronic sinusitis and nasal polyps	IRF4	rs1050979	G/A	imputed DR2=1.00	A	1	+ strand	1 copy of the risk allele
Chronic sinusitis and nasal polyps	LTC4S	rs730012	A/C	chip	C	1	+ strand	1 copy of the risk allele
Allergic rhinitis and mucosal barrier	EMSY/ LRRRC32	rs2155219	G/G	chip	T	0	+ strand	without the risk allele
Allergic rhinitis and mucosal barrier	IL1RL1/ IL18R1	rs10865050	G/G	imputed DR2=1.00	G	2	+ strand	2 copies of the risk allele
Allergic rhinitis and mucosal barrier	TLR1	rs5743604	G/A	imputed DR2=1.00	A	1	+ strand	1 copy of the risk allele
Allergic rhinitis and mucosal barrier	D2HGDH	rs34290285	G/G	chip	G	2	+ strand	2 copies of the risk allele
Allergic rhinitis and mucosal barrier	WDR36/ TSLP	rs1438673	C/T	chip	C	1	+ strand	1 copy of the risk allele
Allergic rhinitis and mucosal barrier	CLEC16A	rs12935657	G/G	imputed DR2=0.98	G	2	+ strand	2 copies of the risk allele
Smell (olfactory receptors)	OR6A2	rs72921001	C/A	chip	A	1	+ strand	1 copy of the effect allele
Smell (olfactory receptors)	OR5A1	rs6591536	A/A	chip	A	2	+ strand	2 copies of the effect allele
Smell (olfactory receptors)	OR7D4	rs61729907	G/A	chip	T	1	complement (– strand)	1 copy of the effect allele
Smell (olfactory receptors)	OR2J3	rs28757581	A/A	imputed DR2=1.00	T	2	complement (– strand)	2 copies of the effect allele
Smell (olfactory receptors)	UGT2A1/ UGT2A2	rs7688383	T/C	imputed DR2=0.98	T	1	+ strand	1 copy of the risk allele